

LAB – 2

Unit, Component & Integration Testing

Objective

Apply three levels of testing:

- I. Unit Testing
- II. Component Testing
- III. Integration Testing

on:

- I. Web Application
- II. Mobile Application
- III. Data Structures & Algorithms Semester Project

Part 1: Unit Testing

Identify at least 5 individual units (functions/methods/classes) in each project and test them independently.

Examples

Web Application

- Login validation function
- Email format checker
- Password encryption function
- API request handler
- Form validation function

Mobile Application

- Input validation
- Button click handler
- Data parsing function
- Local storage function
- API response handler

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- Insert function (e.g., Linked List / Tree)
- Delete function
- Search function
- Sorting function

- Stack/Queue operations

Deliverables

For each unit:

Test Case ID	Related Use Case ID	Test Case Title	Test Objective	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)	Priority	Tested By	Test Date

Minimum: 5 test cases per unit

Part II: Component Testing

Group related units into components/modules and test them.

Examples

Web Application Components

- Authentication Module (Login + Validation + DB Check)
- Product Module (Add/View/Delete)
- Payment Module

Mobile Application Components

- User Profile Module
- API Communication Module
- Notification Module

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- Complete Linked List Module
- Tree Operations Module
- Graph Processing Module

Deliverables

- Component Name
- Included Units
- Test Scenarios
- Observed Behavior
- Bugs Found

Minimum: 3 components per project

Part III: Integration Testing

Test interaction between components/modules

Examples

Web Application

- Frontend ↔ Backend API
- Backend ↔ Database
- Payment Gateway ↔ Order Module

Mobile Application

- App UI ↔ API Server
- Login Module ↔ Dashboard
- Notification Service ↔ User Data

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- Data Input Module ↔ Processing Module
- Algorithm Module ↔ Output Visualization

Required Testing Types

Must apply at least one:

- a) Top-Down Integration
- b) Bottom-Up Integration
- c) Incremental Integration

Deliverables

- Modules Integrated
- Integration Strategy Used
- Test Cases
- Data Passed Between Modules
- Errors/Failures Observed

Minimum: 3 integration scenarios per project

Note: Attaching clear and relevant screenshots for every testing scenario is mandatory. Each screenshot must properly demonstrate the execution of the test case, including inputs, outputs, and system behavior, to provide verifiable evidence of testing. Submissions without complete screenshot documentation for all scenarios will be considered incomplete.